

PRATHAMESH AKOLE

📞 7814000031 📍 Boston, MA ✉ akole.p@northeastern.edu 🌐 [/Prathamesh](#) 🌐 [Portfolio](#)

Available from: Spring 2025

EDUCATION

Northeastern University - Khoury College of Computer Sciences
Master of Science in Computer Science

Sep 2023 – Present
GPA: 3.91 Boston, MA

Savitribai Phule Pune University
Bachelor of Technology in Information Technology

Aug 2019 – Jun 2023
Pune, MH, India

TECHNICAL SKILLS

Languages: Java, C, C++, Python, HTML, CSS, JavaScript
Frameworks: React.js, Node.js, Express.js, Spring Boot, JUnit, Hibernate
Databases: SQL, MySQL, MongoDB, PostgreSQL, DynamoDB, Redis, GraphQL
Cloud & DevOps: Amazon Web Services (AWS), Docker, Kubernetes, CI/CD (Jenkins)
Others: Linux, Bash, RESTful APIs, Microservices, MVC, Git, Agile Methodologies, Test-Driven Development (TDD), Machine Learning, **Computer Vision**,

WORK EXPERIENCE

IDEaS Revenue Solutions

Jan 2023 – Jun 2023

Associate Software Developer Intern

Pune, MH

- Developed and deployed **automated features** using **Python** to improve **system reliability**.
- Engineered **AI-based forecasting tools** using **Python**, **AWS**, and **SQL**; contributed to **computer vision projects** and enhanced **production systems**.
- Optimized complex **SQL DB queries**, reducing **execution time** by **50%** and enhancing **data retrieval efficiency**.

PowerMech Engineering Systems

Software Developer Engineer Intern

Pune, MH

- Engineered a **full-stack** rental platform with **ReactJS**, **NodeJS** and ML features, achieving 100% WCAG compliance.
- Implemented **CI/CD** pipelines reducing deployment time by 60% & increased test coverage to 95%.
- Developed ML tools in **Python**, achieving 85% accuracy in rental forecasts with a self-created **MongoDB** database.
- Wrote 50+ E2E integration tests in Cypress and 80+ Unit tests in JUnit for backend achieving 90% code coverage

PROJECTS

Image Processing Application *Java, MVC, JUnit, SQL*

- * Developed an **Image Processing** Application using **MVC**, Factory, Command patterns, following **SOLID** principles
- * Attained 97% code coverage and minimized potential defects by rigorously writing **JUnit** tests for functionality testing
- * Improved accessibility by integrating images with **SQL** using programming constructs like stored procedures & triggers

Real Time CCTV Violence Detection System Using Deep Learning - *Python, Computer Vision, Data Annotation*

- * Engineered a **computer vision** model with **MobileNetV2** and **LSTM**, in **Python** reducing inference time by 40% with 94% accuracy
- * Built **custom violence detection dataset** from 1000+ video clips across platforms, using **data management** tools.
- * Performed **data annotation** on the entire dataset, labeling **1000+ video clips** for accurate model training.
- * Optimized model architecture to process simultaneous CCTV camera feeds at 30 FPS, reducing false positives by 65%

Real-Time Face Mask Detection System *Python, CNN, Data Annotation*

- * Developed CNN-based mask detection system achieving 98% accuracy across diverse demographics and mask types.
- * Implemented **data pipelines** processing 60+ frames/second, enabling real-time monitoring of multiple video streams.
- * Manually **annotated and labeled** a **custom dataset** of **5000+ images**, combining three unlabeled datasets to ensure diversity in demographics, mask types, and lighting conditions.
- * Enhanced model robustness by training on diverse dataset of 5000+ images, achieving 92% accuracy.

Airport Database Management System - *MySQL, React, Node*

- * Designed a robust application based on database design principles using **MySQL**, ensuring optimal data retrieval
- * Developed a user-accessible interface in **React**, allowing passengers to seamlessly access and view pertinent flight information. Utilized the **Node.js** web framework for backend to enhance system performance and ease API integration

TECHNICAL PUBLICATIONS

- Real-Time CCTV Violence Detection System Using Deep Learning, April 2023, ISBN: 978-1-6654-7524-2.
- Real-Time CNN-Based Face Mask Detection System, December 2022, ISBN: 9781032374598.